The 37th Annual University of Alabama System Applied Mathematics Meeting

University of Alabama in Huntsville

November 1, 2025

SCHEDULE

All talks will be in Room 107, Shelby Center for Science and Technology (SST), while the refreshments and the light lunch will be served in Room 107B, SST. We will have faculty and student discussion in Room 107 and Room 105, SST, respectively. We contacted our Parking Services, and were told that parking permits are not required on Saturday, and you can park wherever space is available on campus.

10:00 Refreshments in Room 107B, SST

10:15-10:25 Welcome remarks by the Department Chair **Dr. Toka Diagana**

10:25-11:05 **Dr. Yiran Wang** (UA) *An Extreme Learning Machine-Based Method for Computational PDEs in Higher Dimensions*

11:05-11:25 **Md Mahmudul Islam** (UAB) *Efficient and Optimally Accurate Ensemble Eddy Viscosity Algorithms for Stochastic Turbulent Flow Problems*

11:25-11:45 **Joshua Vaughan** (UAH) *Efficient* $L1$ *Time-Stepping Schemes for Time Fractional Navier-Stokes Equations*

11:45-2:00 Light lunch in Room 107B, SST

2:00-2:40 **Dr. Summer Atkins** (UAH) *Regularization Techniques for Solving Singular Control Problems in Mathematical Biology*

2:40-3:00 **Trang Ta** (UA) *Long-Term Behavior of Stochastic SIQRS Epidemic Models*

3:00-3:40 **Dr. Andrea Ottolini** (UAB) *Comparative Statics in Optimal Transport*

3:40 Refreshments in Room 107B, SST

4:00-4:30 Faculty Discussion in Room 107, SST

4:00-4:30 Student Discussion in Room 105, SST

ABSTRACTS

Regularization Techniques for Solving Singular Control Problems in Mathematical Biology, *Summer Atkins* (UAH)

Abstract: Optimal control theory has been a promising tool in mathematical biology. In some cases, the problem can have the control appear linearly in both the state equations and in the objective functional. The optimal control for such a problem can have a singular region, a region of nonzero measure for which the switching function (the partial derivative of the Hamiltonian with respect to the control) vanishes identically on that region. Finding explicit solutions for the singular case can be particularly difficult, but what's even more challenging is that numerical solvers are prone to oscillatory artifacts. In this talk, we consider using total variation regularization to regularize these oscillatory artifacts, and we demonstrate numerical solutions to multiple biologically relevant singular control problems.

Efficient and Optimally Accurate Ensemble Eddy Viscosity Algorithms for Stochastic Turbulent Flow Problems, *Md Mahmudul Islam* (UAB)

Abstract: In this talk, we first propose a Reynolds averaged continuous Ensemble Eddy Viscosity (EEV) model for stochastic turbulent flow problems. We then propose a generic algorithm for a family of fully discrete, grad-div regularized, efficient ensemble parameterized schemes for this model. The linearized Implicit-Explicit (IMEX) EEV generic algorithm shares a common coefficient matrix for each realization per time-step, but with different right-hand-side vectors, which reduces the computational cost and memory requirements to the order of solving deterministic flow problems. One family member of the proposed time-stepping algorithm is analyzed and proven to be stable. It is found to be second-order accurate in time for any stable finite element pair. Avoiding the discrete inverse inequality, the optimal convergence of the scheme is proven rigorously for both 2D and 3D problems. For appropriately large grad-div parameters, that scheme is unconditionally stable and allows weakly divergence-free elements. Several numerical tests are given for high expected Reynolds number (E[Re]) problems. The convergence rates are verified using manufactured solutions with E[Re] = 103 , 104 , and 105 . For various high E[Re], the scheme is implemented on benchmark problems which includes: A 2D channel flow over a unit step problem, a non-intrusive Stochastic Collocation Method (SCM) is used to examine the performance of the schemes on a 2D Regularized Lid Driven Cavity (RLDC) problem, and a 3D RLDC problem, and found them perform well. In addition, we propose an efficient penalty projection algorithm for second-order stochastic turbulent-flow problems.

Comparative Statics in Optimal Transport, *Andrea Ottolini* (UAB)

Abstract: Given two probability measures f and g on $\mathbb{R}^m$ and $\mathbb{R}^n$, and a cost function c from $\mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}$, the optimal transport problem is to determine the optimal way to map f to g, where optimality is measures with respect to the cost c. The problem has various applications, and I will focus on an economic one where the map can be interpreted as an assignment of workers to firms. A natural question is that of comparative statics: what is the adjustment in the assignment when the cost varies? In this talk, I will dig more into the economics and the mathematics behind the problem. This is based on joint work with Aleh Tsyvinski and Job Boerma.

Long-Term Behavior of Stochastic SIQRS Epidemic Models, *Trang Ta* (UA)

Abstract: This talk analyzes and classifies the dynamics of SIQRS epidemiological models with susceptible, infected, quarantined, and recovered classes, where the recovered individuals can become reinfected. The models incorporate two important types of random fluctuations. The first type results from small fluctuations of the various model parameters that lead to white noise terms. The second type stems from significant environmental regime shifts that can happen at random. The environment switches randomly between a finite number of environmental states, each with a possibly different disease dynamic. The long-term fate of the disease is fully determined by a real-valued threshold λ. When $\lambda < 0$, the disease goes extinct asymptotically at an exponential rate. On the other hand, if $\lambda > 0$, the disease persists indefinitely. The analysis concludes with some examples in which λ can be computed explicitly, together with simulation results that shed light on real-world situations.

Efficient $L1$ Time-Stepping Schemes for Time Fractional Navier-Stokes Equations, *Joshua Vaughan* (UAH)

Abstract: The Navier–Stokes equations form the cornerstone of viscous fluid flow modeling, extensively studied both theoretically and computationally due to their fundamental role in applications such as aerodynamics, engineering, and plasma physics. Yet, the classical integer-order formulations often fail to accurately capture complex fluid behaviors, such as anomalous diffusion and nonlocal transport in fractal or heterogeneous media. These limitations have motivated the introduction of fractional Navier–Stokes models, incorporating non-integer-order derivatives in time or space to represent memory and nonlocal phenomena more faithfully. In this talk, I will present an $L1$ scheme for the time fractional Navier-Stokes equations and show some theoretical and numerical results.

An Extreme Learning Machine-Based Method for Computational PDEs in Higher Dimensions, *Yiran Wang* (UA)

Abstract: We present two effective methods for solving high-dimensional partial differential equations (PDE) based on randomized neural networks. Motivated by the universal approximation property of this type of networks, both methods extend the extreme learning machine (ELM) approach from low to high dimensions.